

Information Geometry

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1 Geometric Structure of Statistical Models

This note is based on Chapter 2 of [AN00] and [Ama85].

1.1 Statistical Models

$p : \mathcal{X} \rightarrow \mathbb{R}$ is probability function if $p(x) \geq 0$, $\int p = 1$. Statistical Model is parametrized by θ ,

$$S = \{p_\theta = p(x; \theta) | \theta = [\theta^1, \dots, \theta^n] \in \Theta\}.$$

Assume that

- (1) All the $p(x; \theta)$ have a common support.
- (2) $\theta \mapsto p(x; \theta)$ is C^∞ for each $x \in \mathcal{X}$.
- (3) All integral and differentials are exchangeable.

Then, $[\theta^i]$ serves a coordinate system for S . It's tangent space will be $T_\theta = \{A^i \partial_i\}$.

1.2 1-Representation

While $[\theta^i]$ serves a coordinate system, other coordinate systems could exist. In that case those coordinate systems will be related by the Jacobian matrix.

Think of $l(x; \theta) = \log p(x; \theta)$. Then we can build

$$T_\theta^{(1)} = \{A(x) | A(x) = A^i \partial_i l(x; \theta)\},$$

which is a vector space spanned by $\partial_i l(x; \theta)$'s. Moreover, for the coordinate system θ , there exists a *natural isomorphism*

$$\partial_i \in T_\theta \longleftrightarrow \partial_i l(x; \theta) \in T_\theta^{(1)}.$$

This $T_\theta^{(1)}$ is the *1-Representation of the tangent space*. We identify the tangent vector $A^i \partial_i$ to the random variable $A^i \partial_i l(x; \theta)$.

One property is that, since $\mathbb{E}[\partial_i l(x; \theta)] = 0$, for every $A(x) \in T_\theta^{(1)}$, $\mathbb{E}[A(x)] = 0$.

Another property is, if we parametrize statistical model with ξ -coordinate system, then for $A = A^i \partial_i = A^\alpha \partial_\alpha$, they will be related as

$$A^\alpha = B_i^\alpha A^i.$$

Then,

$$A^{(\xi)}(x) = A^\alpha \partial_\alpha l(x; \alpha) = B_i^\alpha A^i \bar{B}_\beta^j \partial_j l(x; \theta) = (B_i^\alpha \bar{B}_\beta^j)(A^i \partial_j l(x; \theta)) = \delta_{ij} A^i \partial_j l(x; \theta) = A^i \partial_i l(x; \theta) = A^{(\theta)}(x).$$

so, *1-Representation is invariant for any coordinate systems*.

1.3 Fisher Metric

Fisher metric of S at the point θ is $G(\theta) = [g_{ij}(\theta)]$, defined by

$$g_{ij}(\theta) = \mathbb{E}_\theta(\partial_i l_\theta \partial_j l_\theta) = \int \partial_i l(x; \theta) \partial_j l(x; \theta) p(x, \theta) dx.$$

This could be also interpreted as natural inner product of 1-representations.

$$\begin{array}{ccc}
\partial_{\theta^i}, \partial_{\theta^j} & \xrightarrow{\text{1-representation}} & A(x) = \partial_{\theta^i} l(x; \theta), B(x) = \partial_{\theta^j} l(x; \theta) \\
\downarrow \text{Fisher metric} & & \downarrow \text{natural inner product} \\
g_{ij}(\theta) = \mathbb{E}_{\theta}(\partial_i l_{\theta} \partial_j l_{\theta}) & \xrightarrow{\text{Equivalent}} & \langle A, B \rangle = \mathbb{E}[A(x)B(x)]
\end{array}$$

Moreover, $g_{ij}(\theta)$ can be computed in different formulas:

$$g_{ij}(\theta) = -\mathbb{E}_{\theta}[\partial_i \partial_j l(x; \theta)].$$

1.4 α -connection

Now, we introduce connections on the statistical model S . The motivation is 1-representation. $\partial_j(\theta + d\theta)$ on $T_{\theta+d\theta}$ have a Taylor expansion

$$\partial_j l(x; \theta + d\theta) = \partial_j l(x; \theta) + \partial_i \partial_j l(x; \theta) d\theta^i.$$

However, since $\mathbb{E}[\partial_i \partial_j l(x; \theta)] = -g_{ij}(\theta) \neq 0$, it does not belong to $T_{\theta+d\theta}^{(1)}$. So correction will be

$$\partial_j l(x; \theta + d\theta) = \partial_j l(x; \theta) + (\partial_i \partial_j l(x; \theta) + g_{ij}(\theta)) d\theta^i.$$

So, it is natural that

$$\Gamma_{ij,k} = \Gamma_{ij}^m g_{mk} = \mathbb{E}_{\theta}[\partial_i \partial_j l(x; \theta) \partial_k l(x; \theta)].$$

Indeed, since

$$\mathbb{E}[\partial_i \partial_j l(x; \theta) + \partial_i l(x; \theta) \partial_j l(x; \theta)] = 0,$$

there are family of connections: α -connections.

$$\Gamma_{ij,k}^{(\alpha)}(\theta) = \mathbb{E}_{\theta} \left[\left(\partial_i \partial_j l(x; \theta) + \frac{1-\alpha}{2} \partial_i l(x; \theta) \partial_j l(x; \theta) \right) (\partial_k l(x; \theta)) \right].$$

Define the 3-tensor

$$\Gamma_{ijk}(\theta) = \mathbb{E}_{\theta}[\partial_i l(x; \theta) \partial_j l(x; \theta) \partial_k l(x; \theta)].$$

then the α -connection is

$$\Gamma_{ij,k}^{(\alpha)} = \Gamma_{ij,k}^{(1)} + \frac{1-\alpha}{2} \Gamma_{ijk}.$$

Theorem 1. *The 0-connection is the Riemannian connection w.r.t. Fisher metric.*

Understanding α -connections via Mixture and Exponential families are helpful.

Example 1 (Mixture Family). *Let $p(x; \theta) = C(x) + \sum_{i=1}^n \theta^i F_i(x)$. Then*

$$\partial_i l(x; \theta) = \frac{F_i(x)}{p(x; \theta)}, \partial_i \partial_j l(x; \theta) = -\frac{F_i(x) F_j(x)}{p(x; \theta)^2}.$$

Therefore, $\partial_i \partial_j l + \partial_i l \partial_j l = 0$. This means $\Gamma_{ij,k}^{(-1)}(\theta) \equiv 0$. Thus, $\nabla^{(-1)}$ is a connection that makes S flat. We call this -1-flat and $\nabla^{(-1)}$ as a Mixture connection or m-connection.

Example 2 (Exponential family). *Let $p(x; \theta) = \exp \{C(x) + \sum_{i=1}^n \theta^i F_i(x) - \psi(\theta)\}$. Then*

$$\partial_i l(x; \theta) = F_i(x) - \partial_i \psi(\theta), \partial_i \partial_j l(x; \theta) = -\partial_i \partial_j \psi(\theta).$$

Therefore, $\Gamma_{ij,k}^{(1)}(\theta) = -\partial_i \partial_j \psi(\theta) \mathbb{E}_{\theta}[F_k(x) - \partial_k \psi(\theta)] = 0$. Thus $\nabla^{(1)}$ is a connection that makes S flat. We call this 1-flat and $\nabla^{(1)}$ as a Exponential connection or e-connection.

Another way to understand this is considering two probabilities $p_1(x), p_2(x)$. Consider the following way to connect two distributions.

$$p(x, t) = (1-t)p_1(x) + tp_2(x).$$

Then

$$\partial_t l(x; t) = \{p_2(x) - p_1(x)\}/p(x; t), \partial_t \partial_t l(x; t) = -\{\partial_t l(x; t)\}^2.$$

Thus, $\nabla^{(-1)}$ connection helps us to understand this connecting as a straight line. We call this -1-straight line. On the other side, consider the following way.

$$p(x, t) = \exp\{(1-t)l_1(x) + tl_2(x) - c(x)\}.$$

Then one can see $\partial_t \partial_t l(x; t) + g_{tt} = 0$. So 1-connection makes this connecting distributions as a straight line.

1.5 Examples

Example 3 (Normal Distribution). *For the parameter $\theta = (\mu, \sigma)$, parameter space is upper half-plane.*

$$\partial_1 = \frac{\partial}{\partial \mu}, \partial_2 = \frac{\partial}{\partial \sigma}.$$

$$l(x; \theta) = -\frac{(x - \mu)^2}{2\sigma^2} - \log(\sqrt{2\pi}\sigma).$$

So 1-representation is

$$T_\theta^{(1)} = \left\{ a \left(\frac{x - \mu}{\sigma^2} \right) + b \left(\frac{(x - \mu)^2}{\sigma^3} - \frac{1}{\sigma} \right) \right\} = \{ ax^2 + bx + c | \mathbb{E}_\theta[ax^2 + bx + c] = 0 \}.$$

Another common parametrization is $\xi = (\mu, \mu^2 + \sigma^2)$, but one could also show that 1-representation are the same.

The Fisher information metric will be calculated as

$$g_{ij}(\theta) = \frac{1}{\sigma^2} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

In ξ coordinate system,

$$g_{\alpha\beta} = \frac{1}{\sigma^4} \begin{pmatrix} 2\mu^2 + \sigma^2 & -\mu \\ -\mu & 1/2 \end{pmatrix}.$$

Now, consider the α -connections. In $\theta = (\mu, \sigma)$ coordinates,

$$\partial_1 \partial_1 l(x; \theta) = -1/\sigma^2, \quad \partial_1 \partial_2 l(x; \theta) = -2 \frac{x - \mu}{\sigma^3}, \quad \partial_2 \partial_2 l(x; \theta) = -\frac{3(x - \mu)^2}{\sigma^4} + \frac{1}{\sigma^2}.$$

Therefore, 1-connection is given by

$$\Gamma_{11,1}^{(1)} = \Gamma_{11,2}^{(1)} = \Gamma_{22,1}^{(1)} = \Gamma_{12,2}^{(1)} = \Gamma_{21,2}^{(1)} = 0$$

$$\Gamma_{12,1}^{(1)} = \Gamma_{21,1}^{(1)} = -\frac{2}{\sigma^3}, \quad \Gamma_{22,2}^{(1)} = -\frac{6}{\sigma^3}.$$

Moreover,

$$\Gamma_{111} = \Gamma_{221} = 0, \quad \Gamma_{112} = \frac{2}{\sigma^3}, \quad \Gamma_{222} = \frac{8}{\sigma^3}.$$

Thus the α -connections are

$$\Gamma_{11,1}^{(\alpha)} = \Gamma_{21,2}^{(\alpha)} = \Gamma_{12,2}^{(\alpha)} = \Gamma_{22,1}^{(\alpha)} = 0, \quad \Gamma_{11,2}^{(\alpha)} = \frac{1 - \alpha}{\sigma^3},$$

$$\Gamma_{12,1}^{(\alpha)} = \Gamma_{21,1}^{(\alpha)} = -\frac{1 + \alpha}{\sigma^3}, \quad \Gamma_{22,2}^{(\alpha)} = -\frac{2 + 4\alpha}{\sigma^3}.$$

From these calculations, one can obtain

$$R_{1212}^{(\alpha)} = \frac{(1 - \alpha^2)}{\sigma^4}.$$

The scalar curvature will be

$$\kappa^{(\alpha)} = -\frac{1 - \alpha^2}{2}.$$

Thus, the normal distribution family is of constant curvature for any α -connections, and especially, ± 1 -flat. The affine coordinate systems are

$$\xi^1 = \mu, \quad \xi^2 = \mu^2 + \sigma^2$$

for -1 -affine coordinates, and

$$\theta^1 = \frac{\mu}{\sigma^2}, \quad \theta^2 = -\frac{1}{2\sigma^2}$$

for 1 -affine coordinates.

Example 4 (Finite $\mathcal{X}$). For $\theta^1 = p_1, \dots, \theta^{n+1} = p_{n+1}$, with $\theta^1 + \dots + \theta^{n+1} = 1$ the statistical manifold is n -dimensional, with

$$p(x; \theta) = \sum_{i=1}^{n+1} \delta(x-i)\theta^i, \quad l(x; \theta) = \sum_{i=1}^{n+1} \delta(x-i) \log \theta^i.$$

So, the 1-representation are

$$T_\theta^{(1)} = \left\{ \sum_{i=1}^n a_i \left(\frac{\delta(x-i)}{\theta^i} - \frac{\delta(x-n-1)}{\theta^{n+1}} \right) \middle| a_i \in \mathbb{R} \right\}.$$

Consequently, the metric tensor in this coordinate system is

$$g_{ij} = \frac{1}{p_i} \delta_{ij} + \frac{1}{p_{n+1}}.$$

Moreover, the α -connections are easily obtained:

$$\Gamma_{ij,k}^{(\alpha)} = \frac{1+\alpha}{2} \left\{ -\frac{1}{(\theta^i)^2} \delta_{i,j,k} + \frac{1}{(\theta^{n+1})^2} \right\}.$$

Theorem 2. The α -connections are all torsion-free. The Riemann curvature tensor is given by

$$R_{ijkm} = (\partial_i \Gamma_{jk}^s - \partial_j \Gamma_{ik}^s) g_{sm} + (\Gamma_{ir,m} \Gamma_{jk}^r - \Gamma_{jr,m} \Gamma_{ik}^r).$$

An exponential family is e -flat, and mixture family is m -flat. Thus, their natural parameters form an affine coordinate system. (Affine coordinate system means, ∂_i become a parallel vector fields.)

1.6 Submanifold (Curved family)

In (S, g, ∇) with coordinate system θ , the submanifold M with coordinate system u inherits Riemannian metric and connection. To be precise, change of coordinate $\iota : M \hookrightarrow S$, $\theta = \theta(u)$ gives Jacobian matrix

$$B_a^i(u) = \frac{\partial \theta^i}{\partial u^a}, \quad i = 1, \dots, n; \quad a = 1, \dots, m.$$

There are no problem in Riemannian metric.

$$g_{ab}(u) = \langle \partial_a, \partial_b \rangle = B_a^i B_b^j g_{ij}(\theta(u)).$$

However, in connection, the problem occurs since $\nabla_{\partial_a} \partial_b$ cannot belong to $T_u(M)$. The $T_u(M)$ components could be calculated as

$$\Gamma_{ab,c} = \langle \nabla_{\partial_a} \partial_b, \partial_c \rangle = (B_a^i B_b^j \Gamma_{ij}^k + \partial_a B_b^k) \langle \partial_k, B_c^m \partial_m \rangle = B_a^i B_b^j B_c^k \Gamma_{ij,k} + (\partial_a B_b^k) B_c^m g_{km}.$$

If we notate the directions that is orthogonal to $T_u(M)$ in $T_u(S)$ by $\{\partial_\kappa\}$, $\kappa = m+1, \dots, n$, then the *Euler-Schouten curvature tensor* or the *imbedding tensor* of M in S is defined by

$$H_{ab,\kappa}(u) = \langle \nabla_{\partial_a} \partial_b, \partial_\kappa \rangle.$$

This is a bilinear mapping $H : \mathcal{T}(M) \times \mathcal{T}(M) \rightarrow (\mathcal{T}(M))^\perp$, sending (A, B) to $(\nabla_A B)^\perp$. Riemann-curvature tensor of M could be calculated as:

$$R_{abcd} = B_a^i B_b^j B_c^k B_d^m R_{ijklm} + g^{\kappa\lambda} (H_{ac,\kappa} H_{bd,\lambda} - H_{bc,\kappa} H_{ad,\lambda}).$$

Example 5. Imagine $M = S^1 \times \mathbb{R} \subset \mathbb{R}^3 = S$ with natural metric and connections. They are both flat, so $R_{abcd} = R_{ijkl} = 0$. However, M is certainly curved in S . H does not vanish! However, by appropriate cancellation, the formula

$$R_{abcd} = B_a^i B_b^j B_c^k B_d^m R_{ijklm} + g^{\kappa\lambda} (H_{ac,\kappa} H_{bd,\lambda} - H_{bc,\kappa} H_{ad,\lambda}).$$

still holds. To summarize, Curvature tensor is intrinsic characteristic of manifold, and does not depend on how they are imbedded. On the other side, Euler-Schouten curvature tensor measures how submanifold is curved in the imbedded structure.

1.6.1 Ancillary submanifolds

One natural formulation of submanifold in statistical model is: for $M \subset S$, attach every $u \in M$ a $A(u)$; $(n-m)$ -dimensional smooth submanifold that transversely meets M at $\theta(u)$. Imagining $A = \{A(u)|u \in M\}$ fills S up smoothly, the pair $\xi = (\xi^\alpha) = (u^a, v^\kappa)$ where $v = (v^\kappa)$, $\kappa = m+1, \dots, n$ gives a coordinate system for S . These $A(u)$ are called *ancillary submanifold rigging* u . In convention, M is set to have $v = 0$ coordinates. The Jacobian matrix of θ and ξ are given by

$$B_\alpha^i = [B_a^i, B_\kappa^i].$$

With these ancillary families,

$$T_u(S) = T_u(M) \oplus T_u(A), \quad u \in M.$$

Naturally, the metric in ξ coordinates decomposes

$$g_{\alpha\beta} = \begin{pmatrix} [g_{\kappa\lambda}] & [g_{\kappa a}] \\ [g_{a\kappa}] & [g_{ab}] \end{pmatrix}.$$

If

$$g_{\alpha\beta} = \begin{pmatrix} [g_{\kappa\lambda}] & 0 \\ 0 & [g_{ab}] \end{pmatrix}.$$

we call $\{A(u)\}$ an orthogonal ancillary family. The connection also decomposes to

$$\Gamma_{\kappa\lambda,\mu} = \langle \nabla_{\partial_\kappa} \partial_\lambda, \partial_\mu \rangle,$$

$$\Gamma_{ab,c}(u) = \langle \nabla_{\partial_a} \partial_b, \partial_c \rangle,$$

and when $\{A(u)\}$ are orthogonal ancillary family,

$$\Gamma_{\kappa\lambda,a}(u) = \langle \nabla_{\partial_\kappa} \partial_\lambda, \partial_a \rangle = H_{\kappa\lambda,a}(u),$$

$$\Gamma_{ab,\kappa}(u) = \langle \nabla_{\partial_a} \partial_b, \partial_\kappa \rangle = H_{ab,\kappa}.$$

Example 6. In the normal distribution $N(\mu, \sigma^2)$, consider the subfamily $N(u, u^2)$, we call these an *Curved Exponential family*.

The model is imbedded in S by $\theta^1 = u, \theta^2 = u$. So $B_a^i = \partial\theta^i/\partial u = 1$ for $i = 1, 2$. $\partial_a = \partial_1 + \partial_2$.

We need to find orthogonal ancillary family. Since

$$g_{ij}(\theta) = \frac{1}{\sigma^2} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix},$$

the $2\partial_1 - \partial_2$ is orthogonal to ∂_a . Thus, attaching

$$A(u) = \{(\theta^1, \theta^2) | 2(\theta^2 - u) + (\theta^1 - u) = 0\}$$

gives it. The coordinate system $\xi = (u, v)$ will be related to θ -coordinate by

$$\theta^1(u, v) = u - 2v, \quad \theta^2(u, v) = u + v.$$

$$B_\alpha^i(u) = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}.$$

Now, the metric tensor will be

$$g_{\alpha\beta}(u, 0) = B_\alpha^i B_\beta^j g_{ij} = \begin{pmatrix} 3/u^2 & 0 \\ 0 & 6/u^2 \end{pmatrix}.$$

α -connections are computed as:

$$\Gamma_{ab,c}^{(\alpha)}(u) = -\frac{7\alpha + 3}{u^3}, \quad \Gamma_{ab,\kappa}^{(\alpha)}(u) = \frac{3 - \alpha}{u^3}, \quad \Gamma_{\kappa\lambda a}^{(\alpha)}(u) = \frac{2(3 - 2\alpha)}{u^3}.$$

So the α -imbedding curvature is $-(\alpha - 3)/u^3$.

2 Dualistic Structure of Statistical Manifolds

This note is based on [Ama85] and [AN00] Chapter 3.

2.1 α -representation

Previously, in Chapter 1.2, we defined 1-representation as a vector space spanned by $\partial_i l(x; \theta)$. To generalize this, we define

$$F_\alpha(p) = \begin{cases} \frac{2}{1-\alpha} p^{(1-\alpha)/2} & \alpha \neq 1, \\ \log p & \alpha = 1. \end{cases}$$

The derivative is $F'_\alpha(p) = p^{-(1+\alpha)/2}$. Now define $l_\alpha(x; \theta) = F_\alpha\{p(x; \theta)\}$ which is α -representation of $p(x; \theta)$. Define

$$T_\theta^{(\alpha)} = \{A(x) | A(x) = A^i \partial_i l_\alpha(x; \theta)\}.$$

This coincides with 1-representation defined earlier. The $T_\theta^{(\alpha)}$ is called the α -representation of the tangent space T_θ . The α -representation of $A = A^i \partial_i \in T_\theta$ will be the random variable

$$A_\alpha(x) = Al_\alpha = A^i \partial_i l_\alpha(x; \theta) = A^i \{p(x; \theta)\}^{(1-\alpha)/2} \partial_i l(x; \theta) = \{p(x; \theta)\}^{(1-\alpha)/2} Al(x; \theta).$$

(Here, we used the formula $\partial_i l_\alpha(x; \theta) = \{p(x; \theta)\}^{(1-\alpha)/2} \partial_i l(x; \theta)$.)

The Fisher information metric was the inner product of $T_\theta^{(1)}$. To define similarly, let α -expectation be

$$\mathbb{E}_\alpha[f(x)] = \int (p(x; \theta))^\alpha f(x) d\mathbb{P}.$$

So then, we have $\langle A, B \rangle = \mathbb{E}_\alpha[(Al_\alpha)(Bl_\alpha)]$.

$$\begin{array}{ccc} A, B \in T_\theta & \xrightarrow{\alpha\text{-representation}} & A_\alpha(x) = Al_\alpha, B_\alpha(x) = Bl_\alpha \\ \downarrow \text{Fisher metric} & & \downarrow \text{natural inner product} \\ \langle A, B \rangle_g & \xrightarrow{\text{Equivalent}} & \mathbb{E}_\alpha[A_\alpha(x) B_\alpha(x)] \end{array}$$

Moreover, the dualistic relation

$$\langle A, B \rangle = \int \{Al_\alpha(x; \theta)\} \{Bl_{-\alpha}(x; \theta)\} d\mathbb{P}.$$

holds from $(\partial_i l_\alpha)(\partial_j l_{-\alpha}) = p(x; \theta)(\partial_i l)(\partial_j l)$.

This dualistic formulation also appears in α -connections. Since

$$\partial_i \partial_j l_\alpha(x; \theta) = \{p(x; \theta)\}^{(1-\alpha)/2} \left(\partial_i \partial_j l + \frac{1-\alpha}{2} \partial_i l \partial_j l \right),$$

$$\Gamma_{ij,k}^{(\alpha)}(\theta) = \int \partial_i \partial_j l_\alpha(x; \theta) \partial_k l_{-\alpha}(x; \theta) d\mathbb{P}.$$

More generally, for three vector fields A, B, C ,

$$\langle \nabla_A^{(\alpha)} B, C \rangle = \mathbb{E}_\alpha[(ABl_\alpha)(Cl_\alpha)] = \int \{ABl_\alpha(x; \theta)\} \{Cl_{-\alpha}(x; \theta)\} d\mathbb{P}.$$

So in α -representation, $\nabla_A^{(\alpha)} B$; α -covariant derivative is just projection of ABl_α to $T_\theta^{(\alpha)}$.

2.2 Dual Affine Connections

We call for (S, g) , ∇ and ∇^* are duals if

$$Z\langle X, Y \rangle = \langle \nabla_Z X, Y \rangle + \langle X, \nabla_Z^* Y \rangle.$$

Equivalent definitions are if $\Pi_\gamma : T_p(X) \rightarrow T_q(S)$ and $\Pi_\gamma^* : T_p(X) \rightarrow T_q(S)$ are parallel transport through curve γ via connections ∇, ∇^* then

$$\langle \Pi_\gamma(X), \Pi_\gamma^*(Y) \rangle_q = \langle X, Y \rangle_p.$$

In coordinate system, it is equivalent to

$$\partial_k g_{ij} = \Gamma_{ki,j} + \Gamma_{kj,i}^*.$$

The constructed α -connections satisfies duality.

Theorem 3. The α - and $-\alpha$ -connections $\nabla^{(\alpha)}$ and $\nabla^{(-\alpha)}$ are mutually dual.

Proof.

$$A\langle B, C \rangle = A \int (Bl_\alpha)(Cl_{-\alpha})d\mathbb{P} = \int (ABl_\alpha)(Cl_{-\alpha})d\mathbb{P} + \int (Bl_\alpha)(ACl_{-\alpha})d\mathbb{P} = \langle \nabla_A^{(\alpha)} B, C \rangle + \langle B, \nabla_A^{(-\alpha)} C \rangle.$$

□

Theorem 4. *When S is flat w.r.t. ∇ , then it is also flat w.r.t. ∇^* .*

Proof. Parallel transport along a loop gives it. □

2.3 α -families, α -affine manifolds

Definition 1. *A family $S = \{p(x; \theta)\}$ of distributions is α -family if their α -representations are written as*

$$l_\alpha(x; \theta) = \theta^i c_i(x) + k(\theta) c_{n+1}(x)$$

with suitable parametrization $\theta = (\theta^1, \dots, \theta^n)$.

This definition first seems abnormal but, after suitable discussions, this definition gains simpler form. First, by letting $\theta^{n+1} = k(\theta)$, the α -representation of α -family can be written as

$$l_\alpha(x; \theta) = \tilde{\theta}^i c_i(x), \quad \tilde{\theta} = (\theta^1, \dots, \theta^{n+1}).$$

Definition 2. *If for some coordinate system θ , $\partial_i \partial_j l_\alpha(x; \theta) = 0$ then we call S an α -affine manifold. The coordinate system is called α -affine coordinate system.*

For α -affine manifold, much simpler description is possible. The l_α should be written as

$$l_\alpha(x; \theta) = C(x) + \sum_{i=1}^n \theta^i F_i(x).$$

Now, return to the α -families, consider the *Denormalization* of $S = \{p_\theta | \theta \in \Theta\}$. Let $\tilde{S} = \{\tau p_\theta | \theta \in \Theta, \tau > 0\}$.

Then the coordinate system $[\theta, \tau]$ could be given to $\tilde{S}$ and every vector field $X = X^i \partial_i \in \mathcal{T}(S)$ could be lifted to $\tilde{X} = X^i \tilde{\partial}_i$. (We notate $\tilde{\partial}_i = \frac{\partial}{\partial \theta^i}$, $\tilde{\partial}_\tau = \frac{\partial}{\partial \tau}$. Define $\tilde{l}_\alpha = F_\alpha(l_\alpha(x; \theta, \tau))$.)

Then following computation holds.

$$\begin{aligned} \tilde{\partial}_i \tilde{l}_\alpha &= \tau^{\frac{1-\alpha}{2}} \partial_i l_\alpha, & \tilde{\partial}_\tau \tilde{l}_\alpha &= \tau^{-\frac{1+\alpha}{2}} p^{\frac{1-\alpha}{2}}, \\ \tilde{\partial}_i \tilde{\partial}_j \tilde{l}_\alpha &= \tau^{\frac{1-\alpha}{2}} \partial_i \partial_j l_\alpha, & \tilde{\partial}_\tau \tilde{\partial}_i \tilde{l}_\alpha &= \frac{1-\alpha}{2} \tau^{-\frac{1+\alpha}{2}} \partial_i l_\alpha, & \tilde{\partial}_\tau \tilde{\partial}_\tau \tilde{l}_\alpha &= -\frac{1+\alpha}{2} \tau^{-\frac{1+\alpha}{2}} p^{\frac{1-\alpha}{2}}. \\ \tilde{g}_{ij} &= \tau g_{ij}, & \tilde{g}_{i\tau} &= 0, & \tilde{g}_{\tau\tau} &= \tau^{-1}. \\ \tilde{\Gamma}_{ij,k}^{(\alpha)} &= \tau \Gamma_{ij,k}^{(\alpha)}, & \tilde{\Gamma}_{ij,\tau}^{(\alpha)} &= -\frac{1+\alpha}{2} g_{ij}, & \tilde{\Gamma}_{i\tau,k}^{(\alpha)} &= \tilde{\Gamma}_{\tau i,k}^{(\alpha)} = \frac{1-\alpha}{2} g_{ik}, & \tilde{\Gamma}_{i\tau,\tau}^{(\alpha)} &= \tilde{\Gamma}_{\tau i,\tau}^{(\alpha)} = 0, & \tilde{\Gamma}_{\tau\tau,k}^{(\alpha)} &= 0, & \tilde{\Gamma}_{\tau\tau,\tau}^{(\alpha)} &= -\frac{1+\alpha}{2} \tau^{-2}. \\ \tilde{\nabla}_X^{(\alpha)} \tilde{Y} &= (\nabla_X^{(\alpha)} Y)^\sim - \frac{1+\alpha}{2} \langle \tilde{X}, \tilde{Y} \rangle \tilde{\partial}_\tau. \\ \tilde{\nabla}_{\tilde{X}}^{(\alpha)} \tilde{\partial}_\tau &= \tilde{\nabla}_{\tilde{\partial}_\tau}^{(\alpha)} (\tilde{X}) = \frac{1-\alpha}{2} \tau^{-1} \tilde{X}. \\ \tilde{\nabla}_{\tilde{\partial}_\tau}^{(\alpha)} \tilde{\partial}_\tau &= -\frac{1+\alpha}{2} \tau^{-1} \tilde{\partial}_\tau. \end{aligned}$$

From these computations, (especially, connections)

Theorem 5. *S is (-1) -autoparallel in $\tilde{S}$. If M a submanifold of S , then the denormalization $\tilde{M}$ satisfies: M is α -autoparallel in S if and only if $\tilde{M}$ is α -autoparallel in $\tilde{S}$.*

Now, the α -families are exactly the one whose denormalization $\tilde{S}$ is α -affine manifold. This is because if $\tilde{S}$ is α -affine manifold,

$$\tilde{l}_\alpha = C(x) + \sum_{\lambda=0}^n \xi^\lambda F_\lambda(x)$$

for appropriate change of coordinates $[\theta^1, \dots, \theta^n, \tau] \longleftrightarrow [\xi^0, \dots, \xi^n]$. Which is exactly the definition of α -family. Moreover, we can fully identify α -families:

First, when $\alpha = 1$, $\tilde{l}_1 = \log \tau + \log p(x; \theta) = C(x) + \sum_{\lambda=0}^n \xi^\lambda F_\lambda(x)$. By letting $\xi^0 = \log \tau + \psi(\xi^1, \dots, \xi^n)$, we get $\log p(x; \xi) = C(x) + \sum_{\lambda=1}^n \xi^\lambda F_\lambda(x) - \psi(\xi^1, \dots, \xi^n)$. This is **exponential family**. For $\alpha \neq 1$ cases, letting $\tau \rightarrow 0$ or ∞ , $\tilde{l}_\alpha \rightarrow 0$ so $C(x) = 0$. Therefore, it should be written as

$$p(x; \theta) = \left(\sum_{i=0}^n \theta^i F_i(x) \right)^{\frac{2}{1-\alpha}}.$$

Typically, $\alpha = -1$ gives

$$p(x; \theta) = \sum_{i=0}^n \theta^i F_i(x),$$

which means (-1) -families are **mixture family**.

Theorem 6. *S be an α -family. Then $M \subset S$ is α -family if and only if M is α -autoparallel in S .*

2.4 Dually Flat Spaces

We have seen: for (S, g) , it is ∇ -flat if and only if ∇^* -flat. This is inherited into submanifolds.

Theorem 7. *Let (S, g, ∇, ∇^*) be dually flat space. If $M \subset S$ is submanifold, autoparallel w.r.t. either ∇ or ∇^* then M is dually flat space w.r.t. $(g_M, \nabla_M, \nabla_M^*)$.*

Proof. Again, we can imagine loop and show that curvature vanishes. $\square$

Remark 1. *Exponential family is (-1) -flat, Mixture family is 1-flat. m -autoparallel submanifold of exponential family is (± 1) -flat. e -autoparallel submanifold of mixture family is (± 1) -flat.*

For the dualistic structure, we can set affine coordinate system $[\theta^i]$ and $[\eta_i]$ which serves for ∇, ∇^* respectively. Then by the definition of dual connection, $\langle \partial_i, \partial^j \rangle$ is constant S . We can thus further choose $\langle \partial_i, \partial^j \rangle = \delta_i^j$. Given dual systems

$$\theta = \theta(\eta), \quad \eta = \eta(\theta)$$

$$p_i = (\partial \eta_k / \partial \theta^i) \partial^k, \quad \partial^j = (\partial \theta^k / \partial \eta_j) \partial_k \text{ so}$$

$$g_{ij} = \langle \partial_i, \partial_j \rangle = \frac{\partial \eta_k}{\partial \theta^i} \langle \partial^k, \partial_j \rangle = \frac{\partial \eta_j}{\partial \theta^i}.$$

Similarly

$$g^{ij} = \frac{\partial \theta^i}{\partial \eta_j}.$$

So $g_{ij} g^{jk} = \delta_i^k$. They are matrix inverse.

Moreover, the dualistic formation produces potential functions. Let $\psi : S \rightarrow \mathbb{R}$ solves the PDE $\partial_i \psi = \eta_i$. Similarly, $\varphi : S \rightarrow \mathbb{R}$ by $\partial^i \varphi = \theta^i$ have a solution. These potentials satisfy

$$\begin{aligned} \varphi + \psi &= \theta^i \eta_i, \\ g_{ij} &= \partial_i \partial_j \psi, \quad g^{ij} = \partial^i \partial^j \varphi, \\ \theta^i &= \partial^i \varphi, \quad \eta_i = \partial_i \psi. \end{aligned}$$

$$\varphi(q) = \max_{p \in S} \{ \theta^i(p) \eta_i(q) - \psi(p) \}, \quad \psi(p) = \max_{q \in S} \{ \theta^i(p) \eta_i(q) - \varphi(q) \}.$$

and are strictly convex.

Example 7. *On the exponential family,*

$$p(x; \theta) = \exp(C(x) + \theta^i c_i(x) - \psi(\theta))$$

is 1-flat as we saw, and the θ -coordinate serves 1-affine coordinate system because

$$\Gamma_{ij,k}^{(1)}(\theta) = \mathbb{E}[(\partial_i \partial_j l)(\partial_k l)] = -\partial_i \partial_j \psi(\theta) \mathbb{E}[\partial_k l] = 0.$$

To find the dual coordinate, since $g_{ij} = -\mathbb{E}[\partial_i \partial_j l(x; \theta)] = \partial_i \partial_j \psi(\theta)$, ψ is the potential function.

$$\eta_i = \partial_i \psi(\theta) = \mathbb{E}[c_i(x)].$$

For this coordinate system will give

$$\Gamma^{(-1)ij,k}(\eta) = 0.$$

Finally, let we find the potential for η -coordinate.

$$H(\eta) = - \int p(x; \theta) \log p(x; \theta) d\mathbb{P} = -\mathbb{E}[\theta^i c_i(x) - \psi(\theta)] = \psi(\theta) - \theta^i \eta_i.$$

Therefore, $\varphi(\eta) = -H(\eta)$ holds.

2.5 Divergences

S be a manifold. $D(\cdot\|\cdot) : S \times S \rightarrow \mathbb{R}$ be smooth, satisfying $D(p\|q) \geq 0$, $D(p\|q) = 0$ if and only if $p = q$. Let we notate

$$\begin{aligned} D((\partial_i)_p\|p') &:= \partial_i D(p\|p') \\ D((\partial_i)_p\|(\partial_j)_{p'}) &:= \partial_i \partial'_j D(p\|p') \\ D((\partial_i \partial_j)_p\|(\partial_k)_{p'}) &= \partial_i \partial_j \partial'_k D(p\|p') \\ D(X_1 \cdots X_l\|\cdot) : p \in S &\rightarrow D((X_1 \cdots X_l)_p\|p) \\ D(\cdot\|Y_1 \cdots Y_l) : p \in S &\rightarrow D(p\|(Y_1 \cdots Y_l)_p) \end{aligned}$$

Then by $D(p\|p) = 0$, $D(p\|q) \geq 0$,

$$D(\partial_i\|\cdot) = D(\cdot\|\partial_i) = 0, \quad D(\partial_i \partial_j\|\cdot) = D(\cdot\|\partial_i \partial_j) = -D(\partial_i\|\partial_j)$$

Divergence on S induces semi-metric

$$g_{ij}^{(D)} := D(\partial_i \partial_j\|\cdot) = D(\cdot\|\partial_i \partial_j) = -D(\partial_i\|\partial_j).$$

If $[g_{ij}^{(D)}]$ is positive definite, we call D a *Divergence*. It also induces connection

$$\Gamma_{ij,k}^{(D)} = -D(\partial_i \partial_j\|\partial_k).$$

In other words,

$$\langle X, Y \rangle^{(D)} = -D(X\|Y), \quad \langle \nabla_X^{(D)} Y, Z \rangle^{(D)} = -D(XY\|Z)$$

Dual divergence is defined $D^*(p\|q) = D(q\|p)$.

Theorem 8. $\nabla^{(D)}$ and $\nabla^{(D^*)}$ are dual w.r.t $g^{(D)}$.

Proof.

$$\partial_k(-D(\partial_i\|\partial_j)) = -D(\partial_k \partial_i\|\partial_j) - D(\partial_i\|\partial_k \partial_j).$$

□

One specific example of divergence is f -divergence. It is defined for f convex on $\{x|x > 0\}$, $f(1) = 0$, strictly convex at $u = 1$.

$$D_f(p\|q) := \int p(x) f\left(\frac{q(x)}{p(x)}\right) d\mathbb{P}.$$

Dual divergence is $f^*(u) = uf(1/u)$, $D_{f^*} = (D_f)^*$. Also, they satisfy monotonicity, for $\kappa = \{\kappa(y|x) \geq 0; x \in \mathcal{X}, y \in \mathcal{Y}\}$ that satisfying $\int \kappa(y|x) = 1$, then $D_f(p\|q) \geq D_f(p_\kappa\|q_\kappa)$.

If we set

$$f^{(\alpha)}(u) = \begin{cases} \frac{4}{1-\alpha^2}(1 - u^{\frac{1+\alpha}{2}}) & \alpha \neq \pm 1 \\ u \log u & \alpha = 1 \\ -\log u & \alpha = -1 \end{cases}$$

then

$$D^{(1)}(q\|p) = D^{(-1)}(p\|q) = \int p(x) \log \frac{p(x)}{q(x)}.$$

so $D^{(-1)}$ is usual KL divergence.

$$D^{(\alpha)}(p\|q) = \frac{4}{1-\alpha^2} \left(1 - \int p(x)^{\frac{1-\alpha}{2}} q(x)^{\frac{1+\alpha}{2}} d\mathbb{P} \right).$$

In particular $D^{(0)}$ is square of the Hellinger Distance.

2.6 Canonical Divergence

For (S, g, ∇, ∇^*) dually flat structure, the *canonical divergence* inducing metric and affine connection exists.

$$D(p\|q) = \psi(p) + \varphi(q) - \theta^i(p)\eta_i(q).$$

To verify that divergence induces metric and connection,

$$D((\partial_i \partial_j)_p \| q) = \partial_i \partial_j \psi(p) = g_{ij}(p),$$

$$\Gamma_{ij,k} = 0$$

which is because dually flatness. Finally, even we transform the coordinate system $\tilde{\theta}^j = A_i^j \theta^i + B^j$, $\tilde{\eta}_j = C_j^i \eta_i + D_j$ then

$$\tilde{\psi} = \psi + D_j \tilde{\theta}^j + c, \quad \tilde{\varphi} = \varphi + B^j \tilde{\eta}_j - B^j D_j - c$$

so divergence is invariant of coordinate system. (As we can see, ψ, φ are **NOT** invariant under affine transformation.)

Theorem 9. Let (S, g, ∇, ∇^*) dually flat structure and $\{[\theta^i], [\eta_j]\}$ be a mutually dual affine coordinate system. D is canonical divergence iff for all $p, q, r \in S$,

$$D(p\|q) + D(q\|r) - D(p\|r) = \{\theta^i(p) - \theta^i(q)\} \{\eta_i(r) - \eta_i(q)\}$$

Theorem 10. For $p, q, r \in S$, γ_1 a ∇ -geodesic of p, q and γ_2 be ∇^* geodesic of q, r . If γ_1, γ_2 are orthogonal, then

$$D(p\|r) = D(p\|q) + D(q\|r).$$

Example 8. For α -families, canonical divergence w.r.t $\nabla^{(\alpha)}$ is exactly the α -divergence defined beforehand.

Theorem 11. The α -divergence in $\pm\alpha$ -family is

$$D_\alpha(p(x; \theta) \| p(x; \theta')) = \mathbb{E}_\theta \left[f_\alpha \left(\frac{p(x; \theta')}{p(x; \theta)} \right) \right].$$

2.7 α -projection

M be a submanifold of S , and let S be a flat manifold with dual coordinate system (θ, η) .

Definition 3. A point $\theta_0 \in M$ is called α -extreme point of $\theta \in S$ when $D_\alpha(\theta, \xi), \xi \in M$ takes an extreme value at $\xi = \theta_0$.

Definition 4. A point $\theta_0 \in M$ is an α -projection of θ to M when α -geodesic connecting θ and θ_0 is orthogonal to M .

If $A_\alpha(\theta_0)$ be the submanifold of S consisting of all the α -geodesics which intersect M at θ_0 perpendicularly. Which is α -autoparallel since it contains all α -geodesics.

To guarantee the α -projection of $\theta \in S$ be unique; we define

Definition 5. A subset $V \subset S$ is α -convex when for any $\theta_1, \theta_2 \in V$, there exists a unique α -geodesic connecting θ_1, θ_2 which is entirely included in V .

Theorem 12. When a closed set $V \subset S$ is $-\alpha$ -convex having smooth boundary ∂V , the α -projection from outside V to the boundary is unique. The unique projection $\theta' \in \partial V$ from θ gives the point which minimizes the α -divergence from θ to V . Especially, the α -projection to a $-\alpha$ -convex submanifold M is unique, giving the α -minimal point.

Proof. Assume we have two points $\theta_1 \neq \theta_2 \in \partial V$ both which are α -extreme points of $\theta \in S - V$ to V . Construct a triangle $(\theta, \theta_1, \theta_2)$ with θ and θ_i ($i = 1, 2$) are connected by α -geodesics and θ_1, θ_2 connected by $-\alpha$ -geodesics. By the Pythagorean theorem,

$$D_\alpha(\theta \| \theta_1) \geq D_\alpha(\theta \| \theta_2) + D_\alpha(\theta_2 \| \theta_1)$$

$$D_\alpha(\theta \| \theta_2) \geq D_\alpha(\theta \| \theta_1) + D_\alpha(\theta_1 \| \theta_2)$$

which gives $D_\alpha(\theta_1 \| \theta_2) + D_\alpha(\theta_2 \| \theta_1) \leq 0$. □

3 Exponential family and the Edgeworth Expansion

3.1 Exponential Family

$$p(x; \theta) = \exp\{x_i \theta^i - \psi(\theta)\}.$$

Here θ is the natural parameter.

$$\partial_i \ell(x, \theta) = x_i - \partial_i \psi(\theta), \quad \partial_i \partial_j \ell(x, \theta) = -\partial_i \partial_j \psi(\theta).$$

Hence

$$\begin{aligned} \mathbb{E}[x_i] &= \partial_i \psi(\theta), & \text{Cov}(x_i, x_j) &= \partial_i \partial_j \psi(\theta), \\ \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)(x_k - \mu_k)] &= \partial_i \partial_j \partial_k \psi(\theta). \end{aligned}$$

Theorem 13.

$$g_{ij}(\theta) = \partial_i \partial_j \psi(\theta), \quad T_{ijk}^{(\alpha)}(\theta) = \frac{1-\alpha}{2} \partial_i \partial_j \partial_k \psi(\theta).$$

The exponential family is 1-flat, and θ is 1-affine. For $\alpha = \pm 1$, the Riemannian and statistical curvatures vanish identically.

The expectation parameter is

$$\eta_i = \partial_i \psi(\theta) = \mathbb{E}[x_i].$$

In the dual coordinates,

$$\begin{aligned} T^{(\alpha)ijk} &= -\frac{1+\alpha}{2} T^{ijk} = -\partial^i \partial^j \partial^k \phi(\eta), \\ \theta^i &= \partial^i \phi(\eta), & g^{ij} &= \partial^i \partial^j \phi(\eta). \end{aligned}$$

The Legendre relation is

$$\psi(\theta) + \phi(\eta) = \theta^i \eta_i.$$

The canonical divergence is

$$D_\psi(\theta_1, \theta_2) = - \int p(x; \theta_1) \log \frac{p(x; \theta_2)}{p(x; \theta_1)} d\mu = \psi(\theta_1) + \phi(\eta_2) - \theta_1^i \eta_{2i}.$$

Example 9. Let

$$S = \{N(\mu, \sigma^2)\}.$$

Then

$$\theta_1 = \frac{\mu}{\sigma^2}, \quad \theta_2 = -\frac{1}{2\sigma^2}, \quad \eta_1 = \mu, \quad \eta_2 = \mu^2 + \sigma^2.$$

The potential functions are

$$\begin{aligned} \phi(\eta) &= -\frac{1}{2} \log(\eta_2 - \eta_1^2) - \frac{1}{2} \log(2\pi e), \\ \psi(\theta) &= -\frac{(\theta^1)^2}{4\theta_2} - \frac{1}{2} \log |-\theta_2| + \frac{1}{2} \log \pi. \end{aligned}$$

The metrics are

$$g_{ij}(\theta) = \begin{bmatrix} \sigma^2 & 2\mu\sigma^2 \\ 2\mu\sigma^2 & 4\mu^2\sigma^2 + 2\sigma^4 \end{bmatrix}, \quad g^{ij}(\eta) = \frac{1}{\sigma^4} \begin{bmatrix} \sigma^2 + 2\mu^2 & -\mu \\ -\mu & 1/2 \end{bmatrix}.$$

The nonzero components of T are

$$T_{111} = 0, \quad T_{112} = 2\sigma^4, \quad T_{122} = 8\mu\sigma^4, \quad T_{222} = 24\mu^2\sigma^4 + 8\sigma^6.$$

Also

$$T_{ijk}^{(\alpha)} = \frac{1-\alpha}{2} T_{ijk}, \quad R_{1212} = (1-\alpha^2)\sigma^6, \quad K = -\frac{1-\alpha^2}{2}.$$

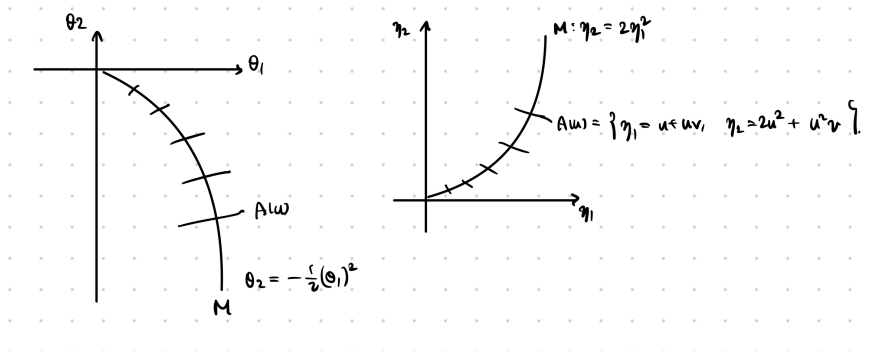


Figure 1: Ancillary manifold

3.2 Curved Exponential Family

$$q(x, u) = \exp\{\theta^i(u)x_i - \psi(\theta(u))\}, \quad u = (u^a), \quad a = 1, \dots, m, \quad \theta = (\theta^i), \quad i = 1, \dots, n.$$

This is an (m, n) -curved exponential family.

Example 10. For $\{N(u, u^2)\}$,

$$\theta_1(u) = \frac{1}{u}, \quad \theta_2(u) = -\frac{1}{2u^2}, \quad \eta_1(u) = u, \quad \eta_2(u) = 2u^2.$$

The coordinates (u, v) can be regarded as a new coordinate system. The ancillary family A is naturally determined from the statistical inference procedure.

Write

$$w^\alpha = w = (u, v) = (u^a, v^\kappa).$$

Then, in θ - and η -coordinates respectively,

$$\partial_\alpha = B_\alpha^i \partial_i, \quad \partial_\alpha = B_{\alpha i} \partial^i.$$

Also,

$$T(S) = T(M) \oplus T_\perp(A), \quad g_{\alpha\beta} = \begin{bmatrix} g_{ab} & g_{a\kappa} \\ g_{\lambda b} & g_{\lambda\kappa} \end{bmatrix}.$$

If $g_{a\kappa} = 0$, then M and $A(u)$ are said to intersect perpendicularly. Hence

$$H_{ab\kappa}^{(\alpha)} = \langle \nabla_{\partial_a}^{(\alpha)} \partial_b, \partial_\kappa \rangle = T_{ab\kappa}^{(\alpha)},$$

which is the α -curvature of M , and

$$H_{\kappa\lambda a}^{(\alpha)} = \langle \nabla_{\partial_\kappa}^{(\alpha)} \partial_\lambda, \partial_a \rangle = T_{\kappa\lambda a}^{(\alpha)},$$

which is the corresponding curvature of $A(u)$. Especially,

$$H_{ab\kappa}^{(e)} = (\partial_a B_b^i) B_{\kappa i}, \quad H_{\kappa\lambda a}^{(m)} = (\partial_\kappa B_{\lambda i}) B_a^i.$$

Example 11. Let

$$A(u) = \{\eta \mid \eta_1 = u + uv, \eta_2 = 2u^2 + u^2v\}.$$

Then

$$B_{\alpha i} = \frac{\partial \eta_i}{\partial w^\alpha} = \begin{bmatrix} 1+v & 2u(2+v) \\ u & u^2 \end{bmatrix}.$$

Moreover,

$$g_{\alpha\beta} = \begin{bmatrix} 3/u^2 & 0 \\ 0 & 3/2 \end{bmatrix},$$

$$T_{abc} = \frac{14}{u^3}, \quad T_{ab\kappa} = -\frac{1}{u^2}, \quad T_{a\kappa\lambda} = \frac{2}{u}, \quad T_{\kappa\lambda\mu} = -4.$$

Therefore

$$H_{ab\kappa}^{(e)} = -\frac{1}{u^2}, \quad H_{\kappa\lambda a}^{(m)} = 0.$$

3.3 Geometrical Aspects of Statistical Inference

Let

$$n_1, \dots, n_N \stackrel{\text{indep}}{\sim} p(x, \theta) \in S.$$

The joint density is

$$\bar{p}(n_1, \dots, n_N; \theta) = \prod_{i=1}^N p(n_i, \theta), \quad \bar{\ell} = \sum_{i=1}^N \ell(n_i, \theta).$$

Thus

$$\partial_j \bar{\ell}(n_1, \dots, n_N; \theta) = \sum_{i=1}^N \partial_j \ell(n_i, \theta).$$

Consequently,

$$\mathbb{E}[\partial_j \bar{\ell} \partial_k \bar{\ell}] = N g_{jk},$$

and

$$\mathbb{E} \left[\partial_j \partial_k \bar{\ell} \partial_m \bar{\ell} + \frac{1-\alpha}{2} \partial_j \bar{\ell} \partial_k \bar{\ell} \partial_m \bar{\ell} \right] = N T_{jkm}^{(\alpha)}.$$

In the exponential family,

$$\bar{p}(n_1, \dots, n_N; \theta) = \exp(N[\theta^j \bar{n}_j - \psi(\theta)]), \quad \bar{n} = \frac{1}{N} \sum_{i=1}^N n_i.$$

Moreover,

$$\mathbb{E}[\bar{n}_i] = \eta_i, \quad \text{Cov}[\bar{n}_i, \bar{n}_j] = \frac{1}{N} g_{ij}.$$

For the MLE,

$$\hat{\eta}_i = \bar{n}_i, \quad \text{Cov}[\hat{\eta}_i, \hat{\eta}_j] = \frac{1}{N} g_{ij}.$$

Thus the Cramer–Rao bound is attained by the MLE.

In a curved exponential family,

$$n_1, \dots, n_N \stackrel{\text{indep}}{\sim} q(x, u).$$

The statistic $\bar{n}$ is attained and is sufficient. There exists $\hat{\eta} = \bar{n}$ in S , but it need not lie in M . The inference map is

$$(\bar{n} = \hat{\eta}) \mapsto \hat{u} = \hat{u}(\bar{n}), \quad \hat{u} : S \rightarrow M.$$

Define

$$A(u) = \{\eta \mid u = \hat{u}(\eta)\}.$$

Assume that this is an $(n - m)$ -dimensional submanifold. Then $A(u)$ is called the ancillary family associated with the estimator.

Remark 2. *The family $A(u)$ does not necessarily pass through $\eta(u)$. There are no such conditions in general.*

Hypothesis testing

Consider

$$H_0 : u = u_0 \quad \text{vs.} \quad H_1 : u \neq u_0.$$

The statistic $\bar{n} = \hat{\eta} \in S$ determines H_0 or H_1 . Let

$$T : S \rightarrow \{r_1, r_2\}, \quad T^{-1}(r) = R.$$

Here R is the critical region; if the observation falls in R , then H_0 is rejected.

3.4 Edgeworth Expansion

Let $A = \{A(u)\}$ be an ancillary family. The observed value is

$$\hat{\eta} = \bar{x} = \eta(\hat{u}, \hat{v}).$$

By the law of large numbers,

$$\bar{x} \rightarrow \eta(u, 0) \quad \text{as } N \rightarrow \infty,$$

assuming that the true distribution is $q(x, u)$. Also

$$\text{Cov } \bar{x} = \frac{1}{N} [g_{ij}(u)] \rightarrow 0.$$

Set

$$\tilde{x} = \sqrt{N}\{\bar{x} - \eta(u, 0)\}, \quad \tilde{u} = \sqrt{N}(\hat{u} - u), \quad \tilde{v} = \sqrt{N}\hat{v}, \quad \tilde{w} = (\tilde{u}, \tilde{v}).$$

Then, expanding near $w = (u, 0)$,

$$\begin{aligned} \bar{x}_i = \eta_i \left(w + \frac{\tilde{w}}{\sqrt{N}} \right) &= \eta_i(u, 0) + \partial_\alpha \eta_i(w) \frac{\tilde{w}^\alpha}{\sqrt{N}} + \frac{1}{2} (\partial_\alpha \partial_\beta \eta_i)(w) \frac{\tilde{w}^\alpha \tilde{w}^\beta}{N} \\ &\quad + \frac{1}{6} (\partial_\alpha \partial_\beta \partial_\gamma \eta_i)(w) \frac{\tilde{w}^\alpha \tilde{w}^\beta \tilde{w}^\gamma}{N\sqrt{N}} + \text{higher order.} \end{aligned}$$

Thus

$$\tilde{x}_i = B_{\alpha i} \tilde{w}^\alpha + \frac{1}{2\sqrt{N}} C_{\alpha\beta i} \tilde{w}^\alpha \tilde{w}^\beta + \frac{1}{6N} D_{\alpha\beta\gamma i} \tilde{w}^\alpha \tilde{w}^\beta \tilde{w}^\gamma + O_p(N^{-3/2}).$$

Equivalently,

$$\tilde{w}^\alpha = g^{\alpha\beta} B_\beta{}^i \tilde{x}_i - \frac{1}{2\sqrt{N}} C_{\beta\gamma}{}^\alpha \tilde{w}^\beta \tilde{w}^\gamma - \frac{1}{6N} D_{\beta\gamma\delta}{}^\alpha \tilde{w}^\beta \tilde{w}^\gamma \tilde{w}^\delta + O_p(N^{-3/2}).$$

Expansion near $(u, 0)!$ $\tilde{w}$

Lemma 1.

$$\begin{aligned} C_{\beta\gamma\alpha} &= T_{\beta\gamma\alpha}^{(m)} = \langle \nabla_{\partial_\beta}^{(m)} \partial_\gamma, \partial_\alpha \rangle, \\ D_{\beta\gamma\delta\alpha} &= \langle \nabla_{\partial_\beta}^{(m)} \nabla_{\partial_\gamma}^{(m)} \partial_\delta, \partial_\alpha \rangle. \end{aligned}$$

Proof sketch. Since

$$\nabla_{\partial_t}^{(m)} d^j = 0,$$

we have

$$\nabla_{\partial_\beta}^{(m)} d^j = 0.$$

Thus

$$\nabla_{\partial_\beta}^{(m)} \partial_\gamma = (\partial_\beta B_{\gamma i}) d^i,$$

and hence

$$\langle \nabla_{\partial_\beta}^{(m)} \partial_\gamma, \partial_\alpha \rangle = B_\alpha{}^i \partial_\beta B_{\gamma i} = B_\alpha{}^i C_{\beta\gamma i} = C_{\beta\gamma\alpha}.$$

A further covariant derivative gives the formula for D . □

The moments are

$$\begin{aligned} \mathbb{E}[\tilde{x}_i] &= 0, \quad \mathbb{E}[\tilde{x}_i \tilde{x}_j] = g_{ij}, \quad \mathbb{E}[\tilde{x}_i \tilde{x}_j \tilde{x}_k] = \frac{1}{\sqrt{N}} T_{ijk}, \\ \mathbb{E}[\tilde{x}_i \tilde{x}_j \tilde{x}_k \tilde{x}_m] &= g_{ij} g_{km} + g_{ik} g_{jm} + g_{im} g_{jk} + \frac{1}{N} S_{ijkl}. \end{aligned}$$

Therefore

$$\tilde{w}^\alpha = g^{\alpha\beta} B_\beta{}^i \tilde{x}_i - \frac{1}{2\sqrt{N}} C_{\beta\gamma}{}^\alpha \tilde{w}^\beta \tilde{w}^\gamma + \dots$$

This gives

$$\mathbb{E}[\tilde{w}^\alpha] = O(N^{-1/2}), \quad \mathbb{E}[\tilde{w}^\alpha \tilde{w}^\beta] = g^{\alpha\beta} + O(N^{-1}),$$

and hence

$$\mathbb{E}[\tilde{w}^\kappa] = -\frac{C^\kappa}{2\sqrt{N}} + O(N^{-1}).$$

Modify $\tilde{w}$ into

$$\tilde{w}^{*\alpha} = \tilde{w}^\alpha + \frac{C^\alpha(\hat{u})}{2\sqrt{N}}, \quad \hat{w}^{*\alpha} = \hat{w}^\alpha + \frac{C^\alpha(\hat{u})}{2N}.$$

Then

$$\mathbb{E}[\hat{w}^*] = w + O(N^{-2}), \quad \mathbb{E}[\tilde{w}^*] = O(N^{-3/2}).$$

Asymptotically,

$$p(\tilde{w}^*; u) = \frac{1}{\sqrt{\det |2\pi g_{\alpha\beta}|}} \exp\left(-\frac{1}{2}g_{\alpha\beta}\tilde{w}^{*\alpha}\tilde{w}^{*\beta}\right) (1 + A_N(\tilde{w}^*)),$$

where

$$A_N(\tilde{w}^*) = \sum_{r=1}^{\infty} N^{-r/2} f_r(\tilde{w}^*)$$

is the Edgeworth expansion. In particular,

$$\begin{aligned} A_N(\tilde{w}^*) &= \frac{1}{6\sqrt{N}} K_{\alpha\beta\gamma} h^{\alpha\beta\gamma}(\tilde{w}^*) + \frac{1}{4N} C_{\alpha\beta}^2 h^{\alpha\beta}(\tilde{w}^*) + \frac{1}{24N} K_{\alpha\beta\gamma\delta} h^{\alpha\beta\gamma\delta}(\tilde{w}^*) \\ &\quad + \frac{1}{72N} K_{\alpha\beta\gamma} K_{\delta\epsilon\sigma} h^{\alpha\beta\gamma\delta\epsilon\sigma}(\tilde{w}^*) + O(N^{-3/2}). \end{aligned}$$

4 Asymptotic Theory of Estimation

4.1 Consistency and Efficiency of Estimators

Let $\hat{u}(\bar{n})$ be a smooth estimator of an (m, m) -curved exponential family

$$M = \{q(x, u)\}.$$

Assume that $\hat{u}$ is independent of N . We regard

$$\hat{u} : S \rightarrow M,$$

and its inverse image is

$$A(u) = \{\eta \in S \mid \hat{u}(\eta) = u\},$$

the ancillary family associated with $\hat{u}$.

Theorem 14. *An estimator $\hat{u}$ is consistent iff, for every $\eta(u)$, the corresponding e.m.c.s. is included in $A(u)$.*

Proof. By the SLLN,

$$\bar{x} \rightarrow \eta(u).$$

□

We use the criterion for determining a good estimator by $\mathbb{E}[\tilde{u}^a \tilde{u}^b]$. Evaluate $\hat{u}$ by

$$\begin{aligned} \mathbb{E}[\tilde{u}^a \tilde{u}^b] &= \mathbb{E}\left[\sqrt{N}(\hat{u}^a - u^a) \cdot \sqrt{N}(\hat{u}^b - u^b)\right] \\ &= g_1^{ab}(u) + g_2^{ab}(u)N^{-1/2} + g_3^{ab}(u)N^{-1} + O(N^{-3/2}). \end{aligned}$$

The estimator $\hat{u}$ is first-order efficient if $g_1^{ab}(u)$ is minimum over all consistent estimators for each u . It is second-order efficient if $g_2^{ab}(u)$ is minimum over all first-order efficient estimators. It is third-order efficient if $g_3^{ab}(u)$ is minimum over all second-order efficient estimators.

Theorem 15. *The covariance of a consistent estimator $\hat{u}$ is*

$$\mathbb{E}[(\hat{u}^a - u^a)(\hat{u}^b - u^b)] = \frac{1}{N} g^{ab} + O(N^{-3/2}),$$

where

$$g^{ab} = (g_{ab} - g_{a\kappa} g_{\lambda b} g^{\kappa\lambda})^{-1}.$$

Therefore a consistent estimator is first-order efficient iff

$$g_{a\kappa} = 0, \quad A(u) \perp M.$$

4.2 Second- and Third-order Efficient Estimator

From the Edgeworth expansion,

$$p(\tilde{w}^*; u) = n(\tilde{w}^*; g_{\alpha\beta})\{1 + A_N(\tilde{w}^*)\}.$$

Recall that

$$\tilde{w}^{*\alpha} = \tilde{w}^\alpha + \frac{C^\alpha(\hat{u})}{2\sqrt{N}},$$

which is bias corrected. Also

$$\begin{aligned} A_N(\tilde{w}^*) &= \frac{1}{6\sqrt{N}} K_{\alpha\beta\gamma} h^{\alpha\beta\gamma}(\hat{w}^*) + \frac{1}{4N} C_{\alpha\beta}^2 h^{\alpha\beta}(\hat{w}^*) + \frac{1}{24N} K_{\alpha\beta\gamma\delta} h^{\alpha\beta\gamma\delta}(\hat{w}^*) \\ &+ \frac{1}{72N} K_{\alpha\beta\gamma} K_{\delta\epsilon\sigma} h^{\alpha\beta\gamma\delta\epsilon\sigma}(\hat{w}^*) + O(N^{-3/2}). \end{aligned}$$

Define a second correction by

$$\hat{w}^{**\alpha} = \hat{w}^{*\alpha} - \frac{(\partial_\kappa C^\alpha) \tilde{v}^\kappa}{2N\sqrt{N}}.$$

This is bias corrected at $(\hat{u}, 0)$. Then

$$p(\tilde{w}^{**}; u) = n(\tilde{w}^{**}; g_{\alpha\beta}) \left\{ 1 + A_N(\tilde{w}^{**}) - \frac{1}{2N} (\partial_\kappa C^\alpha) g_{\alpha\beta} h^{\kappa\beta} + O(N^{-3/2}) \right\}.$$

For an efficient estimator $\hat{u}$,

$$\mathbb{E}[\hat{u}^a - u^a] = b^a(u) + O(N^{-3/2}),$$

where

$$b^a(u) = -\frac{1}{2N} C^a = -\frac{1}{2N} \left(T_{cd}^{(m)a} g^{cd} + H_{\kappa\lambda}^{(m)a} g^{\kappa\lambda} \right).$$

Here $g_{a\kappa} = 0$; the terms correspond to the mixture connection on M and the mixture curvature of A .

Define

$$\begin{aligned} (T^m)_{ab}^2 &= T_{cda}^{(m)} T_{efb}^{(m)} g^{ce} g^{df}, & (T^m)^{2ab} &= g^{ac} g^{bd} (T^m)_{cd}^2, \\ (H_M^e)_{ab}^2 &= H_{ac\kappa}^{(e)} H_{bd\lambda}^{(e)} g^{cd} g^{\kappa\lambda}, & (H_M^e)^{2ab} &= g^{ac} g^{bd} (H_M^e)_{cd}^2, \\ (H_A^m)_{ab}^2 &= H_{\kappa\lambda a}^{(m)} H_{\mu\nu b}^{(m)} g^{\kappa\mu} g^{\lambda\nu}, & (H_A^m)^{2ab} &= g^{ac} g^{bd} (H_A^m)_{cd}^2. \end{aligned}$$

Theorem 16.

$$\mathbb{E}[\tilde{u}^{*a} \tilde{u}^{*b}] = g^{ab} + \frac{1}{2N} \{ (T^m)^{2ab} + 2(H_M^e)^{2ab} + (H_A^m)^{2ab} \} + O(N^{-3/2}).$$

Theorem 17. *From Theorem 5.4, a bias-corrected first-order efficient estimator is automatically second-order efficient, since the $1/\sqrt{N}$ term vanishes. It is third-order efficient if and only if*

$$H_A^m = 0.$$

In this page we specifically consider MLE estimator.

The MLE is the solution of

$$\partial_a \ell(\bar{n}; \hat{u}) = 0.$$

Equivalently,

$$B_a^i(\hat{u}) \{ \bar{n}_i - \eta_i(\hat{u}) \} = 0, \quad B_a^i = \frac{\partial \theta^i}{\partial u^a}.$$

Therefore

$$A(u) = \{ \eta \mid B_a^i(u) (\eta_i - \eta_i(u)) = 0 \}.$$

Hence $A(u)$ is a linear submanifold in η -coordinates, so it is m -flat. Moreover,

$$\langle \partial_a, \partial_\kappa \rangle = 0.$$

Theorem 18. *For the MLE estimator, $A(u)$ is m -flat and $A(u) \perp M$. Therefore it is consistent, first-order efficient, and second-order efficient. The bias-corrected MLE is third-order efficient.*

Example 12. Let

$$M = \{N(u, u^2)\}.$$

The MLE $\hat{u}$ satisfies

$$-\frac{1}{\hat{u}^2}(\bar{n}_1 - \hat{u}) + \frac{1}{\hat{u}^3}(\bar{n}_2 - 2\hat{u}^2) = 0,$$

so

$$\hat{u}^2 + \hat{u}\bar{n}_1 - \bar{n}_2 = 0.$$

The ancillary family is

$$A(u) = \{\eta \mid u\eta_1 - \eta_2 + u^2 = 0\} = (u + uv, 2u^2 + u^2v).$$

For this model,

$$g_{ab} = \frac{3}{u^2}, \quad g_{\kappa\lambda} = \frac{3}{2}, \quad T_{abc}^{(m)} = \frac{4}{u^3}, \quad H_{ab\kappa}^{(e)} = -\frac{1}{u^2}, \quad H_{\kappa\lambda a}^{(m)} = 0.$$

Also

$$b(u) = -\frac{2u}{9N}, \quad \hat{u}^* = \left(1 + \frac{2}{9N}\right)\hat{u}.$$

Moreover,

$$(T^m)^{2ab} = \frac{16}{81}u^2, \quad (H_A^m)^{2ab} = 0, \quad (H_M^e)^{2ab} = \frac{2}{81}u^2.$$

Thus

$$\mathbb{E}[(\hat{u}^* - u)^2] = \left(\frac{1}{3N} + \frac{10}{81N^2}\right)u^2 + O(N^{-3}).$$

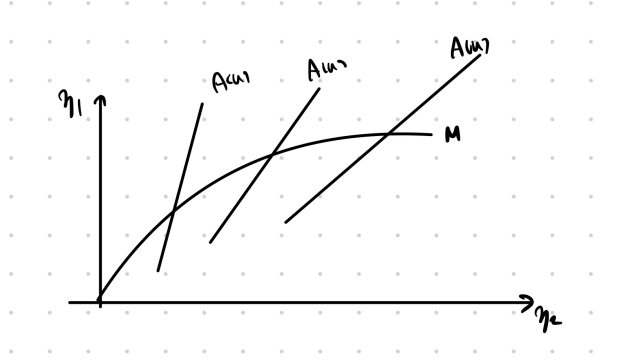


Figure 2: Ancillary manifold

4.3 Third-order Error of Estimator without Bias Correction

Theorem 19.

$$p(\tilde{u}; u) = n[\tilde{u}^i; g_{ab}(u)]\{1 + B_N(\tilde{u}; u)\} + O(N^{-3/2}),$$

where

$$B_N(\tilde{u}; u) = A_N(\tilde{u}^i; u) - \frac{1}{2N}(\partial_a C^d)g_{db}h^{ab}, \quad \tilde{u}^a = \tilde{u}^{*a} + \frac{1}{2\sqrt{N}}C^a(u).$$

Theorem 20.

$$\mathbb{E}[\tilde{u}^a \tilde{u}^b] = \mathbb{E}[\tilde{u}^{*a} \tilde{u}^{*b}] + \frac{1}{4N} [C^a C^b - 2g^{ad}\partial_d C^b - 2g^{bd}\partial_d C^a] + O(N^{-3/2}).$$

Since $\partial_a H_{\kappa\lambda b}^{(m)}$ matters, no uniformly minimizing estimator in u exists. Only the class of bias-corrected estimators ensures that a solution exists.

Example 13. Let $\eta = (y, z)$, where

$$y \sim N(u, 1), \quad z \sim N(0, 1), \quad y \perp z.$$

Let

$$M = \{q(x, u)\}$$

be embedded in

$$p(x; \theta) = \left\{ N_{1,2} \left(\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, I \right) \right\}.$$

Then M is 1-flat in S . In S ,

$$g_{ij} = g^{ij} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad T_{ijk}^{(\alpha)}(\theta) = 0.$$

The most natural estimator is $\bar{y}$. A c -estimator is

$$\hat{u}_c = \bar{y} + c \frac{\bar{z}^2}{\bar{y}}.$$

Its associated ancillary family is

$$A(u) = \left\{ \eta \mid u = \eta_1 + c \frac{\eta_2^2}{\eta_1} \right\},$$

with coordinate $v = \eta_2$. Then

$$\eta_1(u, v) = \frac{1}{2} \left(u + \sqrt{u^2 - 4cv^2} \right), \quad \eta_2(u, v) = v.$$

Furthermore,

$$\mathbb{E}[(\tilde{u}_c^*)^2] = 1 + \frac{2c^2}{u^2} \frac{1}{N} + O(N^{-2}),$$

while

$$\mathbb{E}[\tilde{u}_c^2] = 1 + \frac{c(3c-2)}{u^2} N^{-1} + O(N^{-2}).$$

If $c = 1/3$, then

$$\mathbb{E}[\tilde{u}_{1/3}^2] = 1 - \frac{1}{3u^2} N^{-1} + O(N^{-2}),$$

which is smaller than the MLE value. Thus $\tilde{u}_c$ may be improved over the MLE.

4.4 Ancillary Family Depending on the Number of Observations

$$u \longrightarrow u_N.$$

This depends on N ; this case is omitted here.

4.5 Effects of Parametrization

Geometric features. do not depend on the parametrization of model. $M = \{g(x, u)\}$.

For example, $\hat{u}^*$ is third-order optimal when $A(u) \perp M$ and $A(u)$ is m -flat. But C^a of $\hat{u}$, and the MSE of $\hat{u}^*$, matter in parametrization.

Which parameter system should be used?

1. If $g_{ab}(u)$ is constant, the parameter is a covariance-stabilizing parameter. By Yoshizawa (1991), it exists iff $R_M = 0$.
2. The quantity $(T^m)^{2ab}$ depends on curvature. Can we make this identically zero? This is the minimum covariance parameter; it exists when M is m -flat.
3. Mixture-normal parametrization at u_0 : making the first-order condition locally,

$$T_{abc}^{(m)}(u_0) = 0, \quad \partial_d T_{abc}^{(m)}(u_0) = 0.$$

4. If $g_{ab}(u) = g_{ab}$, modify it to $g_{ab} = \delta^{ab}$. This is covariance-stabilizing and exists only when M is 0-flat. Similarly, locally covariance-stabilizing means

$$T_{abc}^{(0)}(u_0) = 0, \quad \partial_d T_{abc}^{(0)}(u_0) = 0.$$

5. A $-1/3$ -affine parameter satisfies

$$K_{abc} = -3T_{abc}^{(-1/3)}(u) = 0.$$

This is the “zero asymptotic skewness parameter.”

6. A $1/3$ -affine parameter satisfies

$$\mathbb{E}[\partial_a \partial_b \partial_c \ell(x; u)] = 0.$$

Equivalently, $\mathbb{E}[\partial_a \partial_b \ell(x; u)]$ is quadratically approximated. This is the “normal likelihood parametrization.”

Theorem 21. *The α -affine parameters are:*

1. $\alpha = 1$: *locally natural*;
2. $\alpha = 1/3$: *locally normal likelihood parameter*;
3. $\alpha = 0$: *locally covariance-stabilizing*;
4. $\alpha = -1/3$: *locally zero skewness parameter*;
5. $\alpha = -1$: *locally minimum covariance parameter*.

4.6 Geometrical Aspects of Jackknifing

Let

$$x_1, \dots, x_N$$

be independent observations. Let $\hat{u}^{(i)}$ denote the estimator computed from

$$x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_N.$$

Set

$$\hat{w} = (\hat{u}, \hat{v}), \quad \hat{w}^{(i)} = (\hat{u}^{(i)}, \hat{v}^{(i)}), \quad \hat{w}^{(i)} - \hat{w} = \delta \hat{w}^{(i)}.$$

Also write

$$\bar{x}^{(i)} = \bar{x} + \delta \hat{x}^{(i)}.$$

The jackknife estimate is

$$\hat{b} = \frac{N-1}{N} \sum_i \delta \hat{u}^{(i)}.$$

Thus

$$\hat{u}_{JK} = \hat{u} - \frac{N-1}{N} \sum_i \delta \hat{u}^{(i)},$$

and

$$\hat{S} = \frac{N-1}{N} \sum_i \delta \hat{u}^{(i)} \delta \hat{u}^{(i)} - N \hat{b} \hat{b}.$$

Theorem 22. $\hat{u}_{JK}$ coincides with $\hat{u}^*$ up to $O_p(N^{-1})$.

References

- [Ama85] Shun-ichi Amari. *Differential-Geometrical Methods in Statistics*. Vol. 28. Lecture Notes in Statistics. Berlin: Springer-Verlag, 1985. ISBN: 978-3-540-96056-0.
- [AN00] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of information geometry*. American Mathematical Society and Oxford University Press, 2000.